

5.1 Setup

Human salience annotation. We recruited 18 experts across the four domains through our network (3 for *RCT*, and 5 each for *Astro*, *CL*, *QMSum*).⁶ Experts rated the relative salience of each question on a 5-point Likert scale (ranging from 1: least important, to 5: most important). Annotators were asked to motivate their rating through a brief rationale to encourage thoughtful judgments and to allow post-hoc analysis of their decision-making process. To establish a shared understanding between annotators of what content a question may elicit, each question is accompanied by an example answer from a randomly drawn document in the domain. To ensure high annotation quality, we conducted two pilot rounds with four annotators to refine our annotation guidelines (see [Appendix H](#)).

Importantly, the human annotations cannot be regarded as a gold standard for salience. The ratings represent how humans *perceive* question salience, which may not be reflective of how humans actually write summaries. As an initial step toward analyzing human salience through summarization behavior, we explore the application of our framework to human-written summaries in [Appendix E](#).

Model-based salience ratings (LLM-perceived). We prompt LLMs to directly rate question salience. The prompt includes the question list for a given domain and instructions that closely mirror the human annotation guidelines to allow for direct comparison (i.e., 5-point Likert scale and rationales). Each model is prompted 5 times with a shuffled question list to mitigate position bias and to quantify consistency. See [Appendix G](#) for the full prompt.

Analysis method. We use Spearman’s rank correlation coefficient (ρ) to quantify alignment between three measures: human-perceived salience, LLM-perceived salience (both 5-point Likert), and LLM-observed salience (continuous $[0, 1]$).⁷ For groups with multiple ratings, we report averaged pairwise correlation and test for statistical significance with the harmonic-mean p-value ([Wilson, 2019](#)).

Human correlation. Inter-human correlation varies by domain, with meeting summarization

⁶Trained physicians (*RCT*), graduate students/faculty (*Astro*), and graduate students (*CL*, *QMSum*) based in US/Europe.

⁷We take observed salience scores at the 200-words summary length which correlated on average most strongly with human salience. Other scores are explored in [Appendix C](#).

Dataset	Questions	Raters	Spearman	Std.
QMSum	10	5	0.60*	0.18
RCT	21	3	0.46*	0.06
CL	14	5	0.26**	0.29
Astro	13	5	0.16	0.44

Table 2: Inter-annotator correlation for question salience rating. Significance: * ($p < 0.05$) and ** ($p < 0.01$).

(QMSum, $\rho = 0.60$) and RCT abstracts ($\rho = 0.46$) showing a moderate to strong correlation ([Table 2](#)). These domains presumably have established conventions about summary content. In contrast, correlation is weak for summarization of related work (CL, $\rho = 0.26$) and discussion sections (Astro, $\rho = 0.16$). Documents in these domains may vary significantly in the type of content they present (i.e., certain questions may be more relevant to theoretical vs. empirical papers). While our annotation protocol aims to control for this aspect through the example answers by question, there remains annotator subjectivity related to their personal interests.

5.2 Results

To understand if LLMs can reliably rate question salience, we study three conditions. First, as a reference point, we measure consistency of the observational and perceived salience measures estimated over 5 model runs (LLM-observed, LLM-perceived). Second, we study the correlation of LLM-perceived and LLM-observed to measure if models’ explicit ratings align with their summarization behavior (RQ3). Third we correlate LLM-derived salience in human perceived salience (RQ4). We report results for the three conditions in [Table 3](#) and provide qualitative examples in [Table 4](#).

RQ3: When models introspect, does their perceived notion of salience align with their summarization behavior? LLMs have strong and consistent *implicit* notions of salience, but they are unreliable when explicating these preferences in rating tasks. We detail these observations below.

Observational salience is highly stable. We find that observational question salience leads to highly stable scores for all models ($\rho \geq 0.98$). This suggests that LLMs’ underlying summarization process is highly deterministic despite the stochastic nature of language models. Also, it suggests that our proposed approach is a reliable tool for analyzing model behavior.

Models fail to have consistent perceived salience. We find that the consistency of direct

Measure	Random	OLMo	Mixtral	Llama ^{3.1} _{8b}	Llama ^{3.1} _{70b}	4o-mini	4o	Average
Consistency of Saliency Estimates								
LLM-perceived	-0.05	0.20*	0.54**	0.37*	0.71**	0.73**	0.76**	0.47**
LLM-observed	0.92**	0.99**	0.99**	0.98**	0.99**	0.98**	0.98**	0.97**
Correlation of Saliency Estimates								
LLM-perceived vs. -observed	0.03	0.12	0.37*	0.36**	0.47*	0.56**	0.50*	0.35*
Correlation of Model and Human Saliency								
LLM-perceived vs. Human	0.07	0.16	0.41*	0.31*	0.46**	0.51**	0.53**	0.35**
LLM-observed vs. Human	0.20	0.25	0.33*	0.35*	0.36*	0.34*	0.25	0.30*

Table 3: Spearman rank correlations between saliency estimates, averaged across datasets. Per-dataset values in Table 6. Significance: * ($p < 0.05$) and ** ($p < 0.01$); row-wise maxima are bolded.

Question	$L_o/L_p/H$	Rationale LLM-perceived	Rationales Human
(#1, RCT) What significant statistical results are reported?	2.5/4.0/5.0	Statistical results are essential for assessing the validity and reliability of the findings.	($H_{\{1,2,3\}} = 5$) Main outcome that physicians are looking for; very important to be included as measure of success; without it reporting is lacking
(#2, Astro) What broader context or field does the study contribute to or address?	4.6/3.4/2.4	Understanding the broader context helps situate the study within the larger scientific discourse.	($H_{\{2,5\}} = 1$) not relevant, experts would fill context; ($H_1 = 3$) inferable from another question; ($H_3 = 4$) context is important although not always addressed; ($H_4 = 3$) Situating findings within literature, existing models or theories is important
(#3, RCT) What was the main intervention used in the study?	4.2/4.8/5.0	The main intervention is central to understanding the study’s focus and findings.	($H_{\{1,2,3\}} = 5$) determines whether study influences decision making; part of PICO therefore fundamental; essential for communicating purpose/focus of the trial

Table 4: Example questions, saliency scores by LLM-observed (L_o , rescaled to 1-5), LLM-perceived (L_p), humans (H) and summarized rationales. Additional examples in Table 7.

saliency ratings varies significantly for all models and datasets. Generally, strong instruction-following models have more consistent perceived saliency than weaker models (avg. ρ ranges from 0.20 for OLMo to 0.76 for GPT-4o). This finding mirrors recent results in the LLM-as-a-judge literature which demonstrated instability in ratings due to various factors including position bias (Wang et al., 2024; Stureborg et al., 2024).

Perceived $\neq$ observed saliency. Lastly, we find only a weak to moderate correlation between perceived and observed saliency (highest: avg. $\rho = 0.56$ for GPT-4o-mini, lowest: $\rho = 0.12$ for OLMo). Again, stronger instruction-following models show higher correlations, indicating a clear scaling effect. This gap echoes broader findings where generative abilities may not reflect an underlying understanding in models (West et al., 2024).

RQ4: To what extent does model saliency align with human perceived saliency? We find that both LLM-saliency estimates only show a weak to moderate correlation with human saliency perception. Direct rating for question saliency correlates more than observed saliency (highest LLM-perceived: avg. $\rho = 0.53$ for GPT-4o, highest LLM-observed: avg. $\rho = 0.36$ for Llama 3.1

70B). Weak correlation between models and humans holds for all dataset, also those where humans agree more strongly among themselves (Table 6).

In sum, LLM users should carefully consider if a model is appropriate for their summarization task, or provide explicit signals about content priority through prompts or during model training.

6 Related Work

Evaluating and Interpreting Summarization. Recent work suggests that LLMs match or surpass human performance in news summarization (Zhang et al., 2024). However, traditional evaluation protocols remain unreliable especially for LLM-generated summaries (Fabbri et al., 2021; Goyal et al., 2023). This spurred interest in analyzing summarization model behavior. Studies found biases towards content near the beginning/end of documents (Ravaut et al., 2024; Laban et al., 2024). Others analyze training dynamics of summarization models to identify when skills like content selection are learned (Goyal et al., 2022). Extract-then-abstract pipelines (Gehrmann et al., 2018; Li et al., 2021) aim for interpretable text summarization but this interpretability is limited to the document-level (Dhaini et al., 2024). Our research complements prior work by providing a *global in-*

terpretation of what topics LLMs consider important through the lens of text summarization.

Explainable Topic Modeling. Our analysis method draws inspiration from the interpretable topic modeling literature. While classical topic models such as LDA (Blei et al., 2003) have long been used to explain latent themes in text corpora, they are often difficult to interpret (Chang et al., 2009). Recent work showed that LLMs can effectively be used to generate natural language descriptions of latent themes in text mining, clustering and concept induction workflows (Pham et al., 2024; Zhong et al., 2024; Wang et al., 2023; Lam et al., 2024). Our framework uses LLMs to describe salient summary content in form of information-seeking QUDs. The use of QUDs as a representation of information units was shown successful in a wide range of tasks (Newman et al., 2023; Laban et al., 2022; Trienes et al., 2024; Wu et al., 2023b). Finally, in the context of summarization, our work shares theoretical foundations with Wu et al. (2024), who explore human curiosity through inquisitive QUDs. They observe that answering salient questions is a quality indicator for news summaries.

7 Conclusion

We propose an interpretable framework to systematically derive and analyze LLMs’ notion of information salience, a previously latent concept that is nonetheless crucial for text synthesis applications. Our work builds on two key ideas: using length-controlled summarization as a behavioral probe for content selection, and describing what is salient as the answerability of questions. We found that LLMs have a highly consistent notion of salience which is largely compatible across models. We further found that LLMs cannot directly rate the salience of questions, and that model salience weakly aligns with human perceptions. Our work opens new directions to study how LLM salience emerges during training, and for diagnosing content selection challenges in text synthesis tasks.

Limitations

We consider zero-shot prompting with temperature-based decoding to generate summaries. While these settings are common defaults for LLM users, it is conceivable that different prompting styles (e.g., chain-of-density) or decoding methods influence salience patterns. Future work should explore

how these techniques affect salience, particularly in adjacent information-seeking tasks such as query-based summarization.

While our experiments cover diverse disciplines (medicine, astrophysics, computational linguistics, and meetings) and discourse types (structured writing, academic discourse, and dialogue), the texts are primarily technical. Since our framework is designed to be domain-agnostic, we believe it is an exciting direction for future work to explore less technical genres such as fiction (Kim et al., 2024).

Our user study assumed a uniform background and interests among participants, which is a simplification of practical applications. Additionally, the specialized nature of two tasks (i.e., summarizing related work and discussion sections) may have contributed to variability in responses, as even domain experts may not have strong priors on how these texts should be summarized. Future work could explore how differences in expertise and prior knowledge shape perceptions of salience.

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⁸CosmicAI, <https://www.cosmicai.org/>